

# Voice-First LoRa Mesh Networking: Adaptive Spreading Factor and Bandwidth Control with Voice-Aware Routing for Off-Grid Emergency Communication

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September 6, 2026

## Abstract

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LoRa, mesh networking, Codec2, adaptive modulation, emergency communication

## 1 Introduction

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## 2 Literature Review

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## 3 Materials and Methods

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## 4 Results and Discussion

At the time of writing, the custom hardware platform is in the final stage of schematic development and measurement data is not yet available. This section therefore presents the evaluation plan: the metrics that will be measured, the experimental procedure used to obtain them, the baselines against which the system will be compared, and the outcomes that are expected from each experiment.

## 4.1 Evaluation Metrics

The central claim of this work is that a LoRa link can be tuned so that it delivers the highest possible speech quality for the airtime that it consumes. To make this claim measurable, a composite metric, the *voice quality per unit airtime*  $\eta$ , is defined as

$$\eta = \frac{Q}{T_{\text{air}}}, \quad (1)$$

where  $Q$  is the objective speech quality score of the received audio on a five point scale and  $T_{\text{air}}$  is the measured on-air time in seconds required to transmit one second of speech. A value of  $\eta$  greater than the value obtained for a fixed configuration indicates that the adaptive scheme has recovered quality without spending extra airtime.

The following quantities will be logged for every trial.

- **Speech quality  $Q$ .** Objective scoring of the decoded audio against the transmitted reference using ViSQOL, supported by a subjective mean opinion score (MOS) panel of at least fifteen listeners for a representative subset of conditions.
- **Intelligibility.** Word recognition rate obtained from a modified rhyme test, since a compressed speech codec can score poorly on quality yet remain fully understandable.
- **Airtime  $T_{\text{air}}$ .** Computed from the LoRa time-on-air expression for the selected spreading factor, bandwidth, coding rate and payload length, and cross checked against a GPIO toggle captured on a logic analyser at the transmit enable pin.
- **Mouth to ear latency.** Measured end to end with a timestamped audio marker, reported as the median and the ninety-fifth percentile.
- **Packet delivery ratio (PDR) and frame erasure rate,** together with the RSSI and SNR reported by the transceiver for each received frame.
- **Transmissions per delivered frame,** which captures the cost of redundancy in the mesh layer.
- **Channel occupancy,** the fraction of wall clock time the shared channel is busy across all nodes.

## 4.2 Experiment 1: Spreading Factor and Bandwidth Sweep

A two node point to point link will be characterised first, because the mesh results cannot be interpreted without a reliable model of the single hop behaviour. One node transmits a fixed set of speech samples drawn from a standard corpus, encoded at each of the available Codec2 bit rates. The receiving node decodes and stores the audio for offline scoring.

The parameter space is swept exhaustively: spreading factors SF5 through SF12, bandwidths of 203, 406, 812 and 1625 kHz, coding rates of 4/5 through 4/8, and Codec2 modes at 700, 1200, 1600 and 3200 bit/s. Each combination is repeated thirty times at each of five link distances, chosen to span the range

Table 1: Planned test matrix for the link characterisation sweep

| Parameter        | Levels                              |
|------------------|-------------------------------------|
| Spreading factor | SF5 to SF12                         |
| Bandwidth        | 203, 406, 812, 1625 kHz             |
| Coding rate      | 4/5, 4/6, 4/7, 4/8                  |
| Codec2 mode      | 700, 1200, 1600, 3200 bit/s         |
| Link distance    | 5 levels, strong link to link edge  |
| Environment      | Open field, built-up campus, indoor |
| Repetitions      | 30 per cell                         |

from a strong link to the edge of the coverage area. Transmit power is held constant so that only modulation parameters vary. Results are reported as the mean with a ninety-five percent confidence interval.

The outcome of this experiment is a lookup surface of  $\eta$  over the parameter space, indexed by the measured SNR. This surface becomes the decision table used by the adaptive controller.

### 4.3 Experiment 2: Closed Loop Adaptive Control

The adaptive controller selects the spreading factor, bandwidth and codec rate from the surface produced in Experiment 1, using the SNR and frame erasure rate of recent receptions as its input. Its behaviour will be evaluated on a walking test in which one node moves away from a stationary node at a constant pace until the link fails, then returns. Three schemes are compared under the same route and the same speech material:

1. a fixed conservative configuration, high spreading factor and narrow bandwidth, representing a link tuned for worst case range;
2. a fixed aggressive configuration, low spreading factor and wide bandwidth, representing a link tuned for quality;
3. the proposed adaptive controller.

The expected result is that the adaptive scheme matches the aggressive configuration for  $\eta$  while the link is strong, and matches the conservative configuration for PDR once the link degrades, so that the usable communication range is extended without a proportional loss of speech quality. The cost that must be reported honestly is the settling time of the controller, that is, how many seconds of degraded or lost audio occur before the correct configuration is reached after a sudden change in channel conditions.

### 4.4 Experiment 3: Voice-Aware Mesh Routing

A testbed of six to eight nodes will be deployed so that the network diameter is at least three hops. Existing LoRa mesh implementations are text oriented and treat every packet identically, so a managed flooding scheme of the kind used

by Meshtastic is adopted as the baseline. The proposed protocol differs in that it classifies traffic, gives voice frames a short deadline and drops them rather than queuing them once that deadline has passed, and suppresses rebroadcast of a voice frame by a neighbour that is not on a path toward the destination.

Three traffic loads will be applied: voice only, voice mixed with periodic position reports, and a congested load in which two voice streams share the network. For each load the protocols are compared on delivery ratio, ninety-fifth percentile latency, transmissions per delivered frame and channel occupancy.

The expected result is a reduction in transmissions per delivered frame and in tail latency relative to managed flooding, obtained because stale voice frames are discarded instead of consuming airtime that a fresher frame needs. The anticipated trade-off is a lower delivery ratio for non-urgent text traffic during voice activity, which will be quantified rather than avoided.

## 4.5 Threats to Validity

Three risks are recognised. First, objective speech quality metrics are validated mainly on telephone band codecs and may correlate poorly with listener judgement at very low bit rates, which is why the subjective panel is retained. Second, outdoor field results depend on weather, foliage and interference in the shared 2.4 GHz band, so every comparison is run within a single session and interference is logged with a spectrum capture. Third, a testbed of eight nodes cannot demonstrate behaviour at the scale of a large deployment, so the measured per hop behaviour will be used to parameterise a simulation for the larger case, and the simulation will be reported as an extrapolation rather than as a measurement.

## 5 Conclusion

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## Acknowledgment

## References

- [1] Placeholder reference.